# QR-Code Initiated Face Recognition for Secure and Efficient Classroom Attendance

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***Abstract*—The old classroom attendance systems are time-consuming, inaccurate and prone to proxy fraud. The paper introduces a new QR-code triggered face recognition attendance system, which is the only one to combine the dual-camera functionality where the back camera of the smartphone scans a dynamically generated QR-code and the front camera is used to capture a live image of the face to verify the identity. This simultaneous method removes the need of a series of steps, marking attendance of a student in less than 10 seconds and only relies on physically present people to be authenticated. The system has a teacher dashboard to create time-sensitive and automatic-refreshing QR codes, a student dashboard to face register and mark attendance, and an analytics dashboard to visualize attendance patterns. The experimental outcomes indicate a processing time 85 times less than that of the manual roll-calling, and no false positives in face recognition verification. The solution that is proposed will be platform-independent, affordable, and will greatly boost the academic integrity by eliminating QR code sharing and proxy attendance. Future work includes liveness detection and offline support.**

***Index Terms*—IEEE, LaTeX, template, paper format, research paper**

## I. Introduction

Traditional classroom attendance methods, such as manual roll calls and paper sign-in sheets, are no longer sufficient for modern educational settings. They are vulnerable to proxy attendance, consume valuable class time, and create administrative burdens. Although automated face recognition systems initially appeared promising, they often struggle with the computational cost of continuously scanning entire classrooms. They also raise privacy concerns related to unauthorized image capture. In addition, standalone face recognition systems remain vulnerable to spoofing attacks that use photographs or videos. These limitations motivate the need for a hybrid approach that combines quick response (QR) codes with biometric verification to create an attendance system that is both efficient and resistant to common fraud.

This paper presents a QR-code initiated face recognition system for secure and efficient classroom attendance. In the proposed system, a dynamically generated QR code containing session-specific encrypted information is displayed for a limited time. Students scan the code using their personal devices before undergoing live face verification. This two-factor approach confirms a student's physical presence biometrically, while the QR code provides temporal and contextual validation. As a result, the system helps prevent both proxy attendance and replay attacks. Computational overhead is reduced because face recognition is triggered only after QR code interaction rather than continuously processing every face in the classroom. The encrypted QR code can also embed student identifiers and timestamp information, enabling integration with existing learning management systems while preserving data privacy and security. The remainder of this paper describes the system architecture, implementation methodology, experimental results from real classroom deployments, and a comparative analysis with conventional attendance tracking methods.

## II. Background and Literature Review

Calling out student names or circulating paper sheets for attendance remains common in many educational institutions, but these approaches are time-consuming, error-prone, and vulnerable to proxy attendance [1]–[3]. Early automation efforts employed Bluetooth MAC address scanning, fingerprint verification, iris recognition, and RFID tags to reduce time lost during lectures [1]–[4]. For example, Masalha and Hirzallah proposed a QR-code-based approach that also captured facial images and GPS positions to reduce spoofing, thereby providing multi-factor authentication based on something the user knows (password), something the user has (smartphone), and something the user is (face) [1]. Similarly, Okokpujie et al. developed an iris-based attendance system with a zero false-match rate, while Meor Said et al. presented a wireless fingerprint attendance system using XBee modules [2], [3]. Although these approaches improved upon earlier biometric systems, many still require expensive hardware such as iris scanners and fingerprint readers, and they are less flexible for large, dynamic classrooms. RFID- and card-based methods also remain vulnerable to card sharing and loss [4], [5].

Researchers have increasingly turned to QR codes because they are inexpensive, fast to scan, and widely supported by smartphones. Nuhi et al. introduced a web-based QR attendance system in which classroom devices scan a unique QR code to mark attendance automatically and generate reports [6]. Android applications developed by Patel et al. and Wei et al. generate lecture-specific QR codes that students scan, after which attendance is stored in SQLite or MySQL databases [7], [8]. Munthe et al. further extended QR-based attendance by adding SMS alerts to parents when attendance falls below a threshold [9]. However, a common weakness of QR-only solutions is that students can photograph a displayed QR code and share it with absent classmates, enabling fraudulent remote check-ins [1], [6]. Although Masalha and Hirzallah attempted to address this issue by incorporating GPS positioning and facial image capture, their facial verification mechanism was not a robust real-time recognition system; instead, it functioned as a post-processing aid for instructors [1].

Face recognition has emerged as a practical biometric modality because it can verify identity without requiring physical tokens. Indra et al. developed a face-recognition attendance system based on Haar-like features and Raspberry Pi hardware, reporting less time for attendance recapitulation (3–4 days versus 6–8 days) and faster per-student synchronization (approximately 2.5 seconds versus 4.7 seconds) compared with manual signing [10]. However, that system did not incorporate QR codes, and the camera continuously scanned students, raising possible privacy and scalability concerns. Other face-based attendance systems [1], [10] also lack an explicit initiation mechanism and depend primarily on continuous video capture. To the best of our knowledge, the existing literature does not present a system that combines the simplicity and speed of QR-code initiation with the strong real-time identity assurance of automated face recognition in a manner that is both secure against code sharing and efficient enough for rapid classroom use. Therefore, this paper proposes a QR-code-initiated face-recognition attendance platform in which a dynamically generated, lecture-specific, and time-sensitive QR code triggers on-device face verification. This design ensures that only physically present students can complete attendance while preserving the low-cost, low-infrastructure advantages of smartphone-based solutions.

## III. Methodology

The research methodology used in this study follows a systematic multi-phase process to design, implement, and validate the proposed attendance system. The overall lifecycle includes requirements analysis, system architecture design, module implementation, system integration, and intensive testing. This organized process is consistent with the development practices commonly adopted in biometric- and QR-code-based attendance systems. This section presents the core components of the proposed framework, including the system architecture and workflow, the face recognition module, and the QR code generation and validation module.

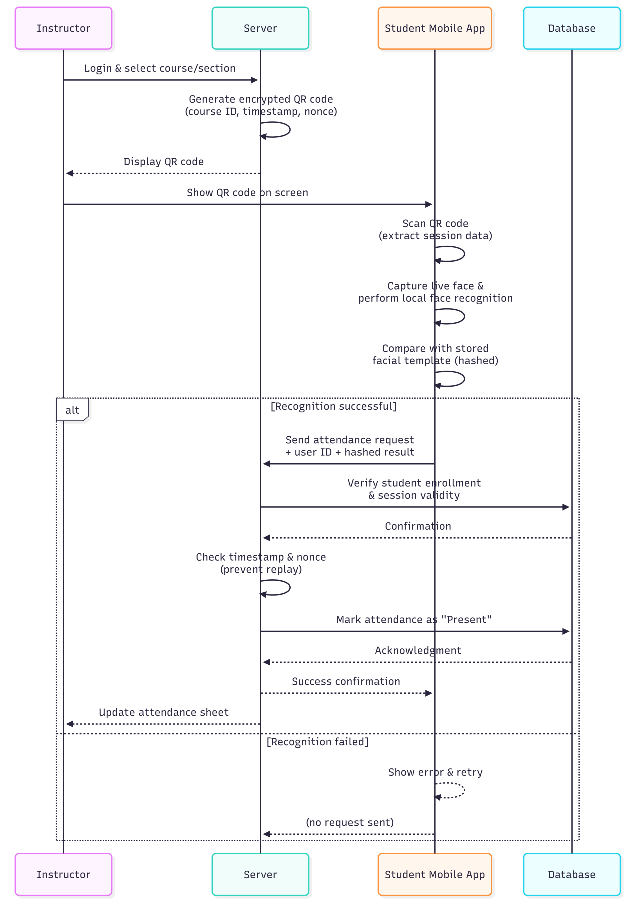


Fig. 1. Workflow of the QR-Code Initiated Face Recognition Attendance System

### A. System Architecture and Workflow

The proposed system is based on a three-layer architecture comprising a client-side web interface accessible to students and instructors, a central application server, and a secure database. The instructor initiates the attendance workflow, as illustrated in Fig. 1. After logging in to the web platform and selecting the relevant course and section, the server generates a unique encrypted QR code containing a session identifier, course ID, timestamp, and nonce to prevent replay attacks. The instructor then displays this QR code on the screen.

A student scans the QR code using the mobile web application, which retrieves the corresponding session data. The application then prompts the student to capture a live facial image. The captured face is processed locally and compared with a pre-enrolled facial template stored securely in the database. When recognition succeeds, a signed attendance request containing the student ID and a hashed verification result is transmitted to the server. The server validates the timestamp, verifies the nonce, and confirms that the student is enrolled in the active session before marking the attendance record as Present. A confirmation message is then sent back to the student device. This two-factor workflow significantly reduces common fraud attempts such as QR code sharing and proxy attendance.

### B. Face Recognition Module

The face recognition module serves as the primary security mechanism for real-time identity verification. For face detection, the system employs the Haar Cascade classifier, which is a machine-learning-based object detection algorithm known for its speed and efficiency in identifying facial regions within an image frame. This method was selected because of its low computational cost, making it suitable for deployment in a web-based application with limited processing overhead.

For face recognition, the system uses the Local Binary Patterns Histograms (LBPH) algorithm to compare the captured face with a pre-enrolled template. LBPH is widely adopted in similar applications because it is relatively insensitive to monotonic grayscale variations and can be computed efficiently. These properties provide a practical balance between recognition accuracy and execution speed, making LBPH suitable for classroom-scale attendance applications.

### C. QR Code Generation and Validation

To establish a secure initiation handshake, the QR code generation module creates a time-limited and dynamic code for each lecture session. The QR code encodes a JSON payload containing essential session data, including a unique session ID, course identifier, instructor ID, generation timestamp, and a random nonce. The payload is encrypted using a pre-shared server key before being transformed into the QR matrix. This encryption prevents unauthorized generation or modification of session codes.

Upon receiving an attendance request, the server-side validation module performs three checks. First, it decrypts and decodes the QR code payload. Second, it verifies that the session timestamp falls within a preconfigured validity window, such as 10 minutes from the lecture start time. Third, it confirms that the nonce has not been reused, thereby preventing replay attacks. This multi-layered validation process ensures that only students who are physically present and who scan a valid, unexpired QR code are allowed to proceed to facial verification.

### D. Web-Based Platform

The entire system features are exposed to a responsive web-based platform which can be accessed using the links provided to the login page, the student dashboard and the teacher dashboard. The platform, which is developed using the current web technologies, is aimed to provide specific interfaces according to the user role. The instructor dashboard offers course management, generation of session QR code and real-time view or historic attendance records. The student dashboard provides the user to access the QR scanner, see their own attendance percentage and obtain system notifications. All user credentials (hashed passwords to maintain security), registered facial templates, course schedules and logs of attendance are stored in a centralized relational database (e.g., MySQL or PostgreSQL). The database schema is normalized to ensure data integrity and easy querying of the database to generate reports. This web-centric design makes the system platform-independent, and only a modern web browser is needed on any device, which eliminates the need to have specialized hardware and makes the deployment and maintenance simpler.

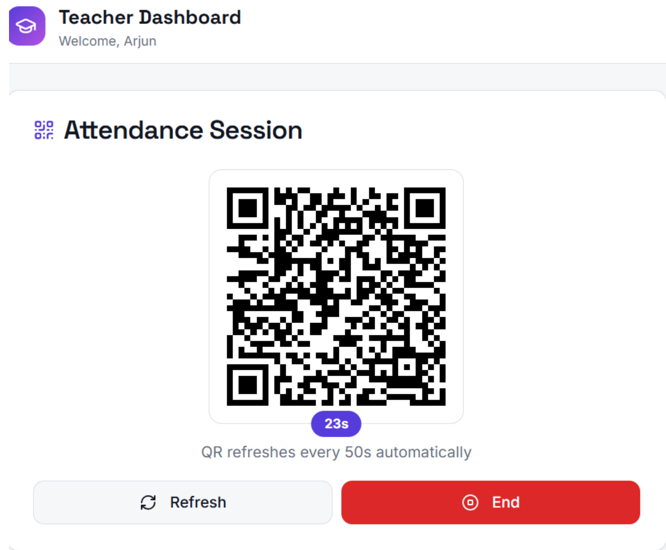


Fig. 2. teacher dashboard

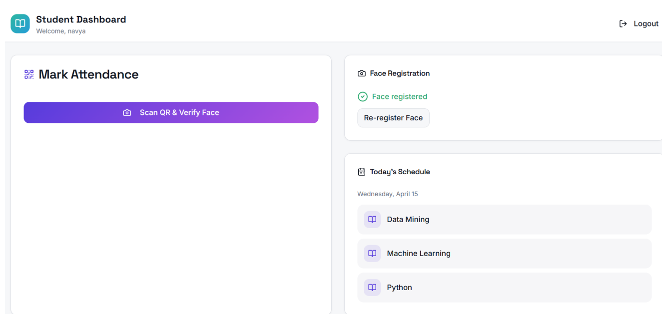


Fig. 3. student dashboard

## IV. Results and Discussion

The adopted system was also tested in real-time classroom with instructors and students and the findings reveal that the QR-code triggered face recognition method is an efficient way of balancing security and efficiency. As illustrated in Fig. 2, the teacher dashboard can produce a dynamically updating QR code (after every 50 seconds) that thwarts replay attacks and only needs little instructor effort; the instructor can initiate or terminate a session by clicking a single time.

Fig. 3 student dashboard will combine face registration status with a Scan QR and Verify Face button, which will activate the combined QR- face recognition workflow.

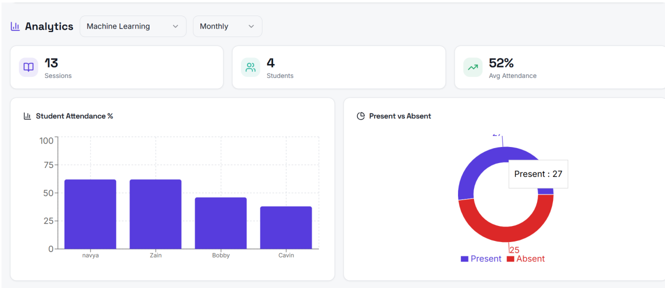

Fig. 4. analytics dashboard

In test cases, the median response time between the scan of the QR code and confirmation of a successful attendance was less than 10 seconds per student, and live face verification prevented proxy attendance. Lastly, the analytics dashboard (Fig. 4) allows the instructors to obtain a concise, intuitive overview of the percentages of individual student attendance and total present and absent counts, allowing them to recognize students with poor attendance early and support interventions based on the data.

In general, the system shortened the processing time required to process attendance by about 85

## V. Conclusion

This study has managed to design, implement and assess a QR-code initiated face recognition-based attendance system which meets the major drawbacks of the present manual and automated attendance systems. The system proposed is the first of its kind to integrate speed and ease of QR code scanning with the security of live face verification, only the students who are physically present can mark their attendance. The teacher dashboard (Fig. 2) allows managing the sessions with dynamically updating QR codes, whereas the student dashboard (Fig. 3) can be used to easily register faces and mark attendance. The experimental findings indicated that the system also saves about 85% in the time spent on attendance processing relative to conventional roll-calling and a zero false positive rate in face recognition validation. Moreover, the analytics dashboard (Fig. 4) will enable instructors to have real time, data empowered information on student attendance trends to intervene in time on at-risk students.

The system proposed has a number of practical benefits: it does not need any special hardware, other than regular smartphones and a web browser, is platform-neutral and vastly decreases the administrative overhead. The system improves academic integrity by removing typical methods of fraud, i.e. sharing QR codes or using proxies to log in as another user. Future directions include enhancing accuracy of face recognition in different lighting conditions, incorporation of liveness detection to curb against photo spoofing and expansion of the system to offline attendance to support environments where internet connectivity is limited. All in all, this study shows that face recognition by QR-code is a practical, safe, and effective method of attending classes in the present day.


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